Nuclear Fuel Performance

NE-533

Spring 2026

Exam 1

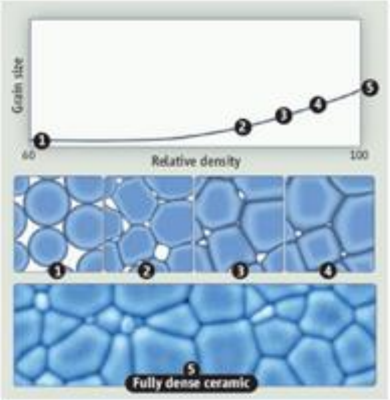
- Sorry for the delay, had to wait on EOL to get exams to me
- Avg. 83, stdev 19
- Applied a 7-point curve on top of your raw scores to get to an average of 90
- Solutions posted on moodle
- Let me know if you have any issues with my grading
- Come and get help if you need it

Last Time

- Talked through numerical thermomechanics
- All fuel performance codes: a) Numerically model the temperature in the fuel, b) Numerically model the stress in the cladding, c) Consider gap pressure, closure, and heat transfer
- Some multiphysics fuel performance codes
  - FRAPCON/FRAPTRAN, BISON, OFFBEAT, etc.
- Intro into material property evolution
  - the microstructure changes during the lifetime of the fuel
  - impacts properties and phenomena
- Materials 101 on structure, point defects, etc.

Material processing

- The processes we use to make a material have a huge impact on its microstructure and properties
- **Casting** – manufacturing process in which a liquid material is poured into a mold and then allowed to solidify
  - Can be used to make complex shapes
  - The solidified microstructure typically has properties that are far from ideal
- **Sintering** – Forming a solid from a powder using heat and/or pressure without melting the material
  - Applicable to metals and ceramics
  - Difficult to obtain a material that is fully dense
  - Used to make fuel pellets
- **Post-fabrication processing**
  - **heat treatment**
  - **working**

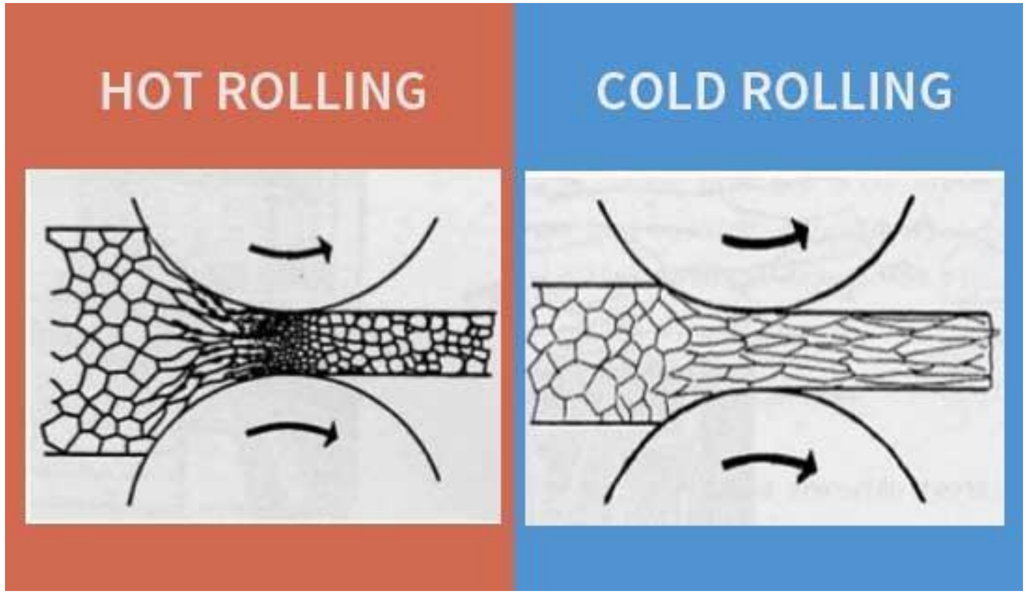


Material processing

- Heat treatment: heating or cooling a material to extreme temperatures to get desired microstructure and properties
  - Used to control the rate of microstructure change, including diffusion, grain growth, or phase change
  - Annealing
    - heat and hold the material, which allows for any defects in the material to repair themselves
    - The metal is then allowed to cool back to room temperature at a slow pace to produce more ductile material
  - Quenching
    - heated to a temperature that transforms its internal structure, held at that temperature, then rapidly cooled
    - The quick cooling process locks in the high T structure, or forces phase transformation without the ability to relax stresses

Material processing

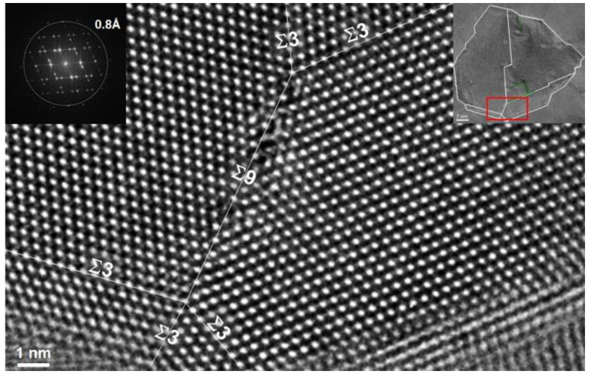
- Working: metalworking process that plastically deforms the alloy
- Cold working occurs below the recrystallization temperature, hot working occurs above
- By plastically deforming the material, we are increasing the number of dislocations and dislocation barriers, increasing the yield strength
- Main types of working processes:
  - Squeezing (rolling/extrusion/drawing)
  - Bending (shaping)
  - Shearing (slitting)

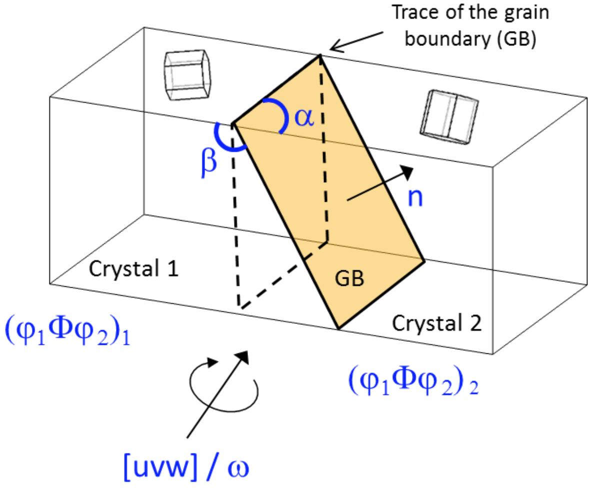


GRAIN PROPERTIES/BEHAVIOR

Grain Boundaries

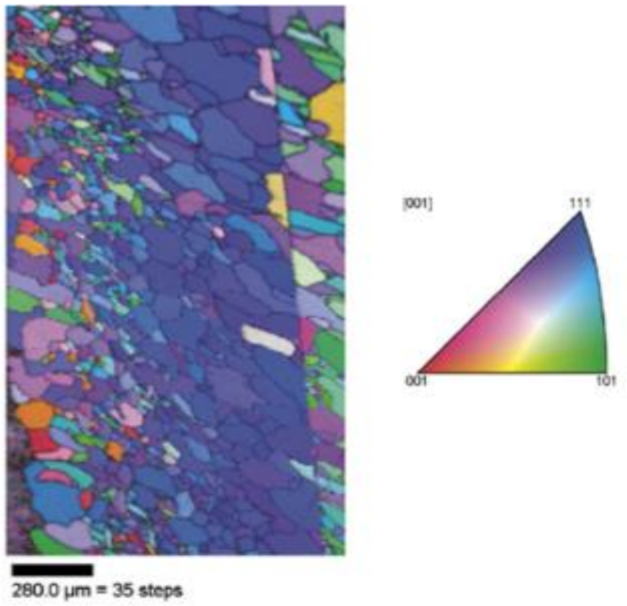
- Materials are typically composed of various regions where the crystal lattice is oriented differently
- When two grains meet, there is a plane of atoms that do not follow the crystal lattice called a **grain boundary**
- Grain boundaries add energy to the material that is a function of their structure
- A grain boundary's energy is determined by its:
  - **Inclination** – the orientation of the 2D grain boundary plane (2 degrees of freedom)
  - **misorientation** – the rotation required to align one grain with the other (3 degrees of freedom)



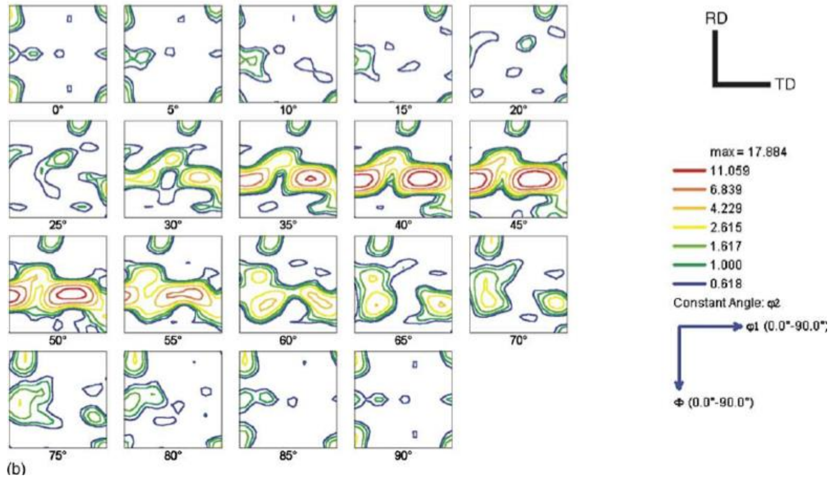


Grain Texture

- The orientation of all the grains in a material is called the texture
- The degree of texture is dependent on the percentage of crystals having a preferred orientation
  - randomly oriented grains equals no texture
- Texture is seen in almost all engineered materials, and can have a great influence on materials properties



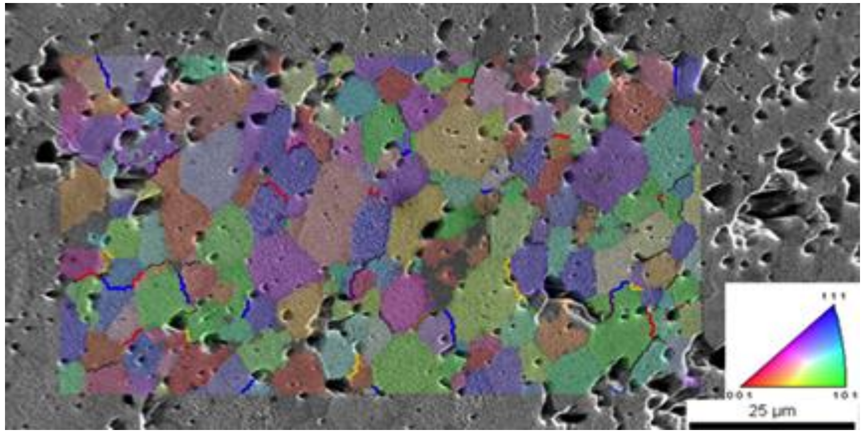
Electron back scatter diffraction map for cubic crystal structure



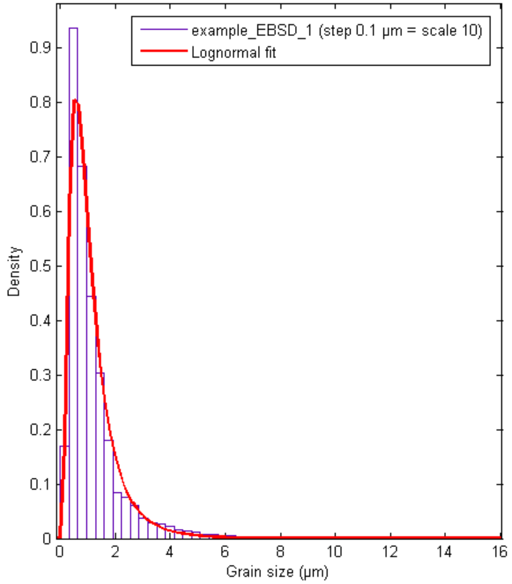
Orientation distribution function

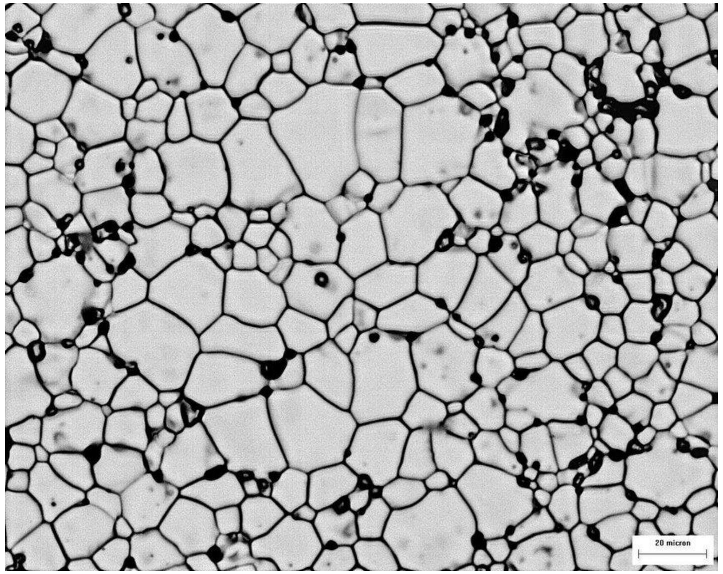
Grain size

- In a polycrystal, there is also a distribution of grain sizes
- The average grain size of the fuel has a significant impact on its behavior
- Typical LWR fuel has an initial average grain size of about 10 microns
- The average grain size impacts
  - Fission gas release
  - Swelling
  - Thermal conductivity
  - Creep



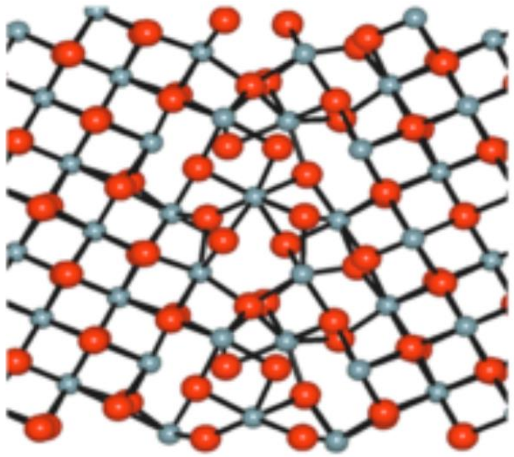
UO2 grain structures

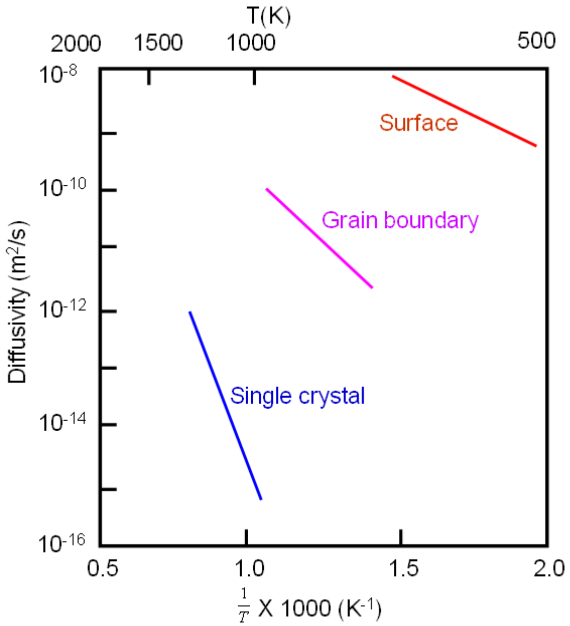




Grain boundary diffusion

- Diffusion often occurs faster along grain boundaries than in the perfect crystal
- Grain boundaries have more space than the perfect lattice
- So, atoms diffuse faster along grain boundaries than through the perfect lattice
- This means that grain boundary, or intergranular, diffusion increasingly dominates bulk (intragranular) diffusion
- This can become more pronounced as the temperature is reduced
- Grain boundary diffusion has a large impact on creep

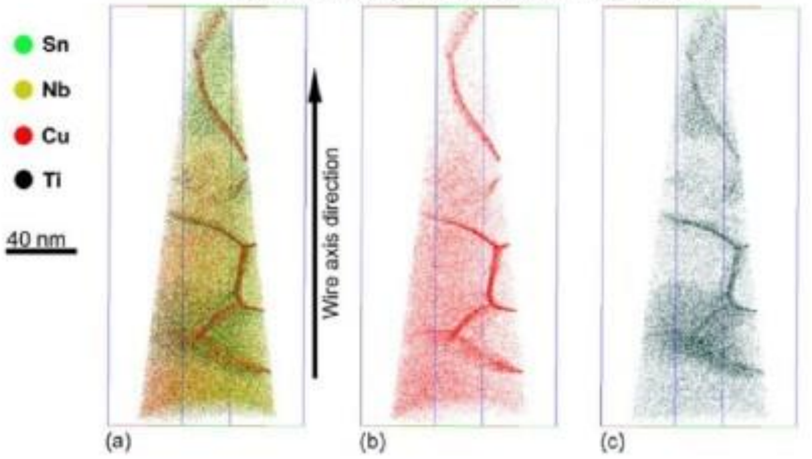




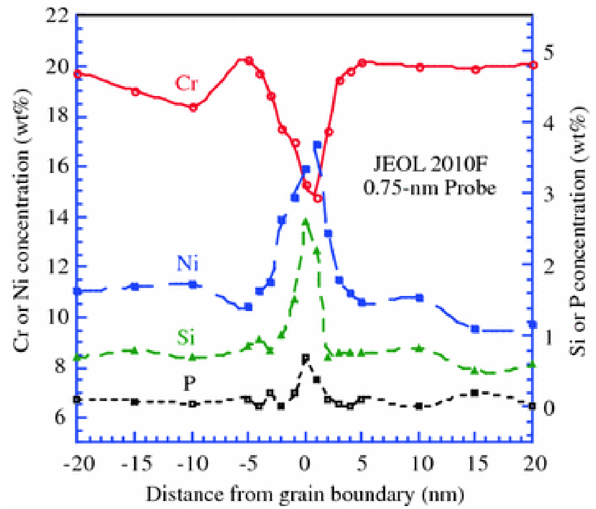
Grain boundary segregation

- Impurity atoms (such as fission gas) and other defects move to grain boundaries
- Due to grain boundaries having more open space, impurity atoms have a lower energy when they are on grain boundaries
- This is called **grain boundary segregation**
- Radiation induced segregation (RIS) is where radiation produces defects that drive towards the grain boundary and preferentially enrich or deplete solute atoms at the grain boundary

M J R Sandim Supercond. Sci. Technol. **26** (2013) 055008

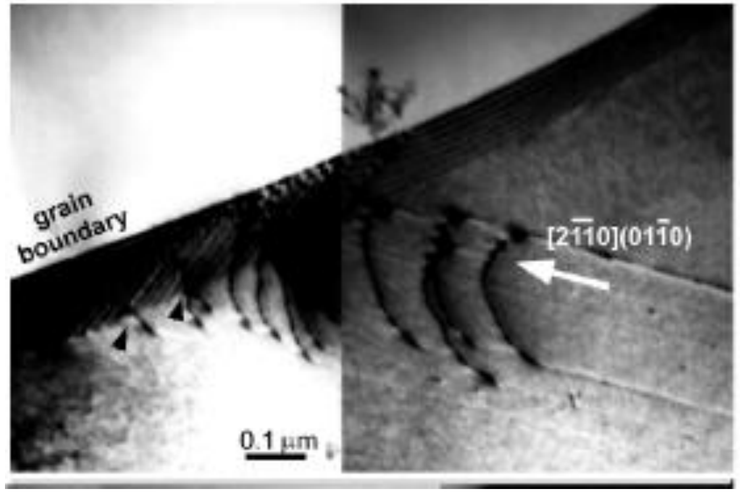


APT reconstruction : (a) elements mapped in the sample and (b) and (c) show only Cu and Ti

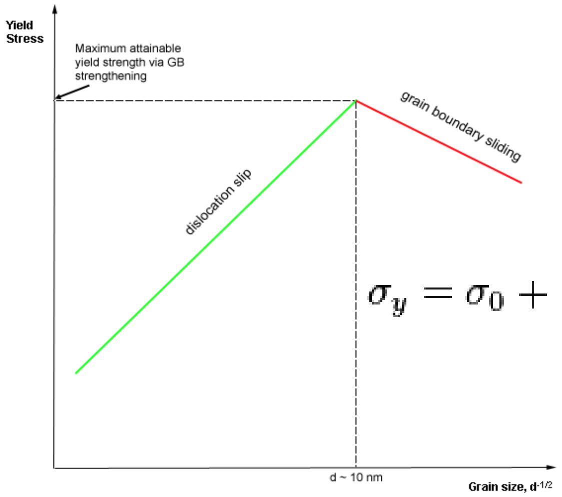


Dislocation interaction with grain boundaries

- Grain boundaries impede dislocation motion
- The number of dislocations within a grain impacts how easily dislocations can traverse grain boundaries and travel from grain to grain
- So, by changing grain size one changes barrier density, influencing dislocation movement and yield strength
- This is called the Hall-Petch effect
  - where σ_y is the yield stress, σ_0 is a materials constant for the starting stress for dislocation movement (or the resistance of the lattice to dislocation motion), k_y is the strengthening coefficient (a constant specific to each material), and d is the average grain diameter



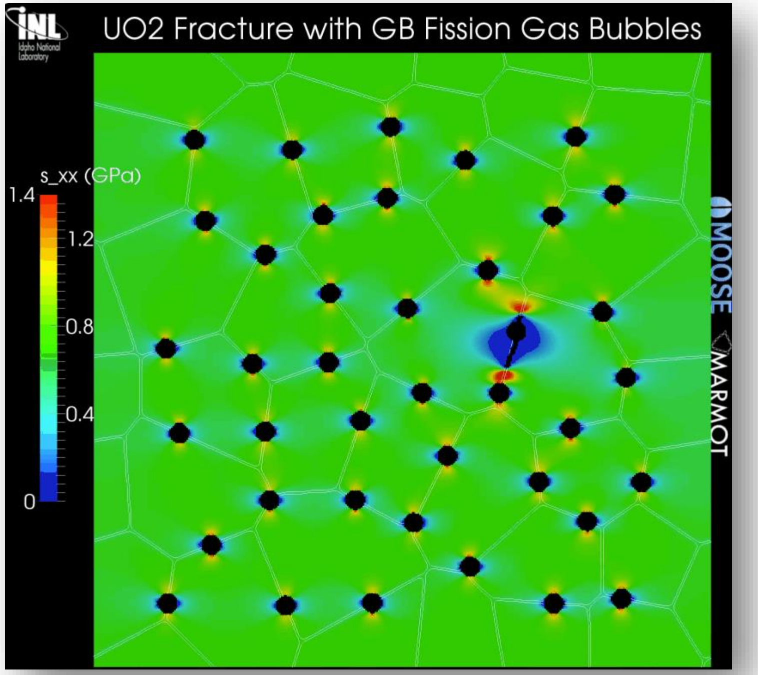
Hall-Petch Strengthening Limit

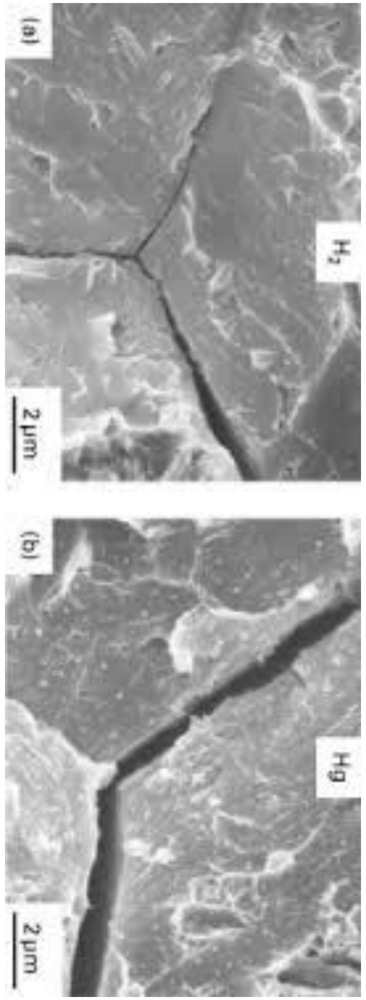


$$\sigma_y = \sigma_0 + \frac{k_y}{\sqrt{d}}$$

Grain boundaries and cracks

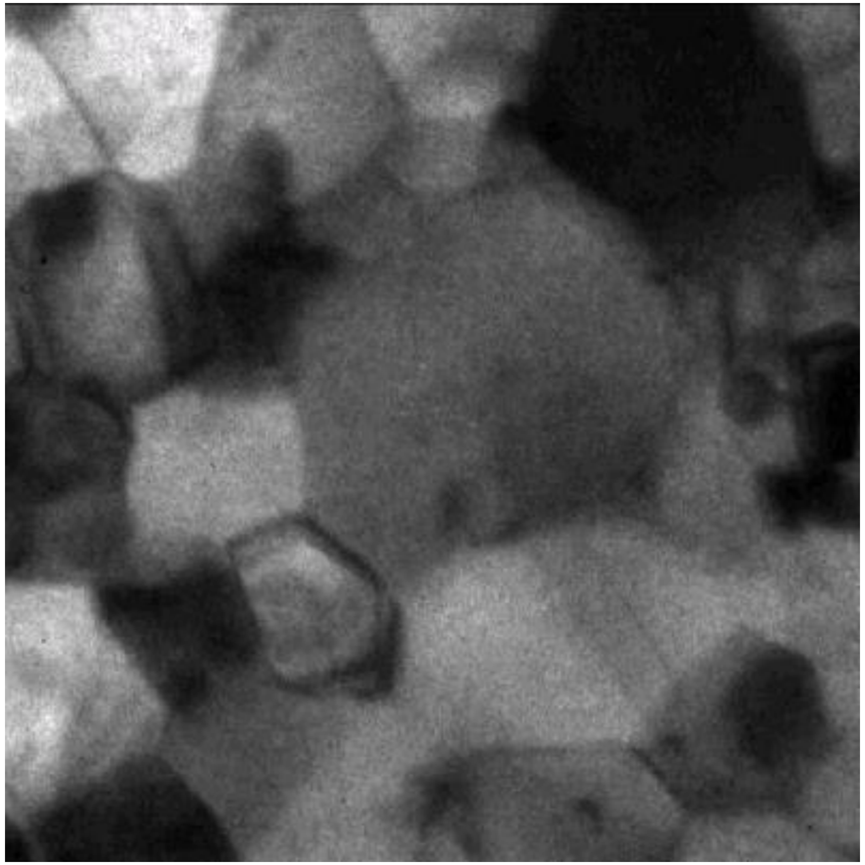
- Materials often crack along grain boundaries
- Grain boundaries can be weaker than the perfect lattice
  - different bonding environment, increased extrinsic particles, etc.
- Thus, fracture will often occur along grain boundaries (intergranular fracture)





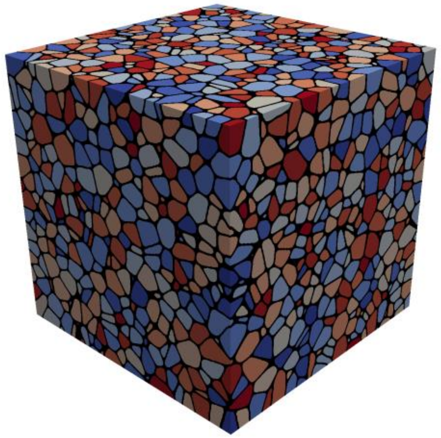
Grain size change

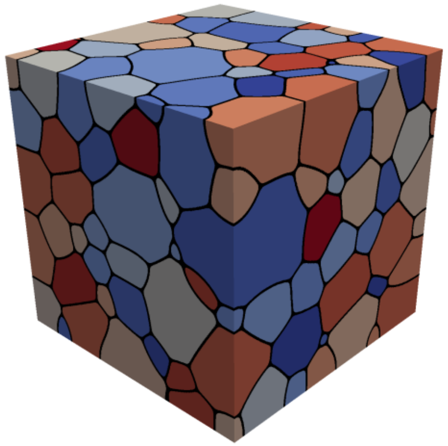
- A single crystal has a lower energy state than a polycrystal, due to the grain boundary energy
- So, grain boundaries migrate to reduce the energy
- $v_{GB} = M_{GB}\,(P_d - P_r)$
  - M_{GB} is the grain boundary mobility
  - P_d is the driving force (pressure) for grain growth
  - P_r is the pressure resisting grain growth

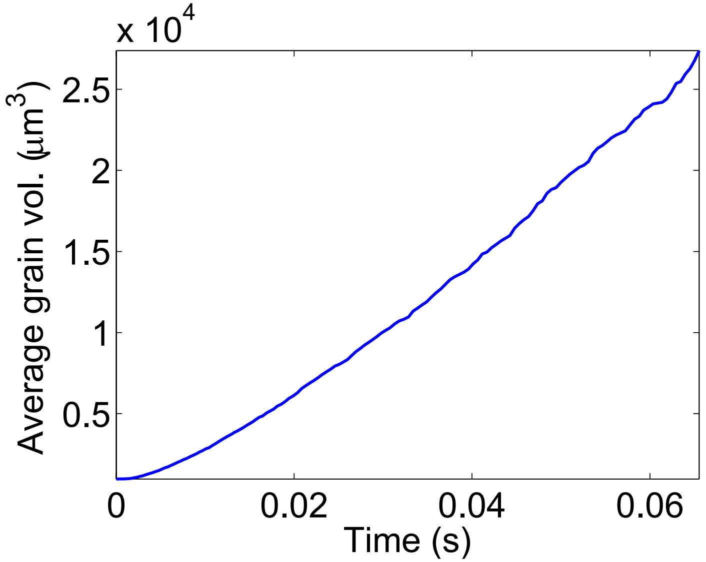


Grain growth

- Due to grain boundary migration, the average grain size goes up with time during annealing/heat treating
- As grain boundaries migrate, some grains grow and some shrink
- Shrinking grains eventually disappear
- The average grain volume $\bar{V}_{gr} = V_{mat}/N_{grains}$
- Therefore, as N_{grains} decreases due to grain disappearance, the average grain volume goes up







Grain boundary mobility

- The grain boundary mobility is a function of temperature and GrB type
- The grain boundary mobility is determined according to
  - $M_{GB} = M_0 e^{-\frac{Q}{k_b T}}$ m^4/(J s)
  - k_b = 8.6173e-5 eV/K is the Boltzmann constant
- Both the pre-factor M_0 and the activation energy Q change as a function of the grain boundary misorientation
- We often use an average grain boundary for a material, taken from polycrystal measurements-> UO2: M_0 (m^4 /J-s) = 4.6e-09, Q (eV) = 2.77
- What is the mobility of UO_2 at 1600 K?
  - M_{GB} = 4.6e-9*exp(-2.77/(1600*8.6173303e-5)= 8.7e-18 m^4/J-s

Grain growth

- There are various driving forces for grain growth
- The most common driving force is the reduction of grain boundary energy
  - $P_d = \frac{2\gamma_{GB}}{R}$, where γ_{GB} is the GB energy and R is the radius of curvature
  - It is often called the curvature driving force, because it drives grain boundaries to be straight
  - It also causes larger grains to grow at the expense of smaller ones
- Other driving forces include
  - Temperature gradients, Elastic energy gradients, Dislocation energy gradients
- Velocity of a spherical grain: $v = M\frac{2\gamma_{GB}}{R}$
- Simplified size (diameter) of a spherical grain vs time: $D(t)^2 - D_0^2 = 2M\gamma_{GB}t$

Maximum grain size

- Consider a material with an average grain size D
- The change in D can be written as

$$\frac{dD}{dt} = k\left(\frac{1}{D} - \frac{1}{D_m}\right)$$

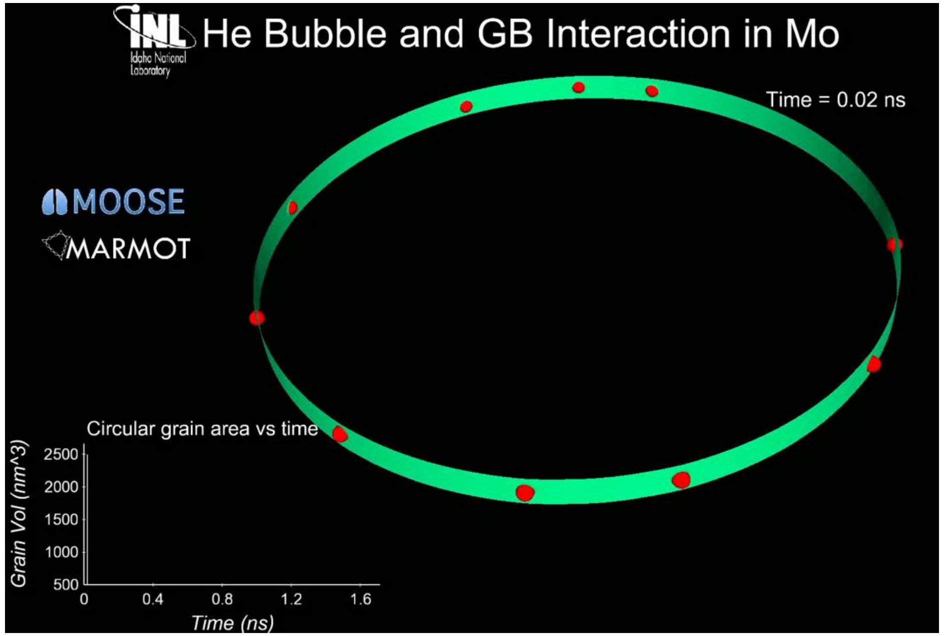
  - $k = 2\, M_{GB}\, \gamma_{GB}$ is a rate constant that can be determined from experiments
  - D_m is the grain size at which the driving force equals the resistive pressure
- For UO2:

Material	M_0 (m^4 J/s)	Q (eV)	γ_{GB} (J/m^2)
UO_2	4.6e-09	2.77	1.58

- D_m is a function of temperature $D_m = 2.23 \cdot 10^3\ \exp(-7620/T)$ microns

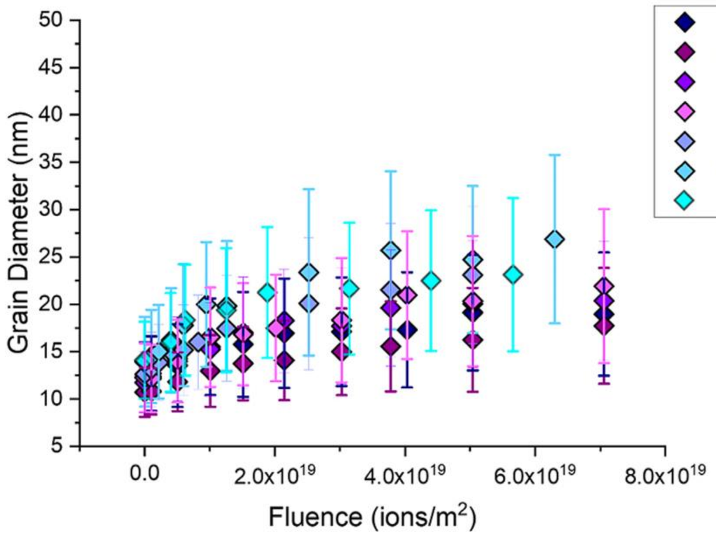
Grain growth

- Grain boundary motion is inhibited by pores, precipitates, solute atoms, etc.
- Solute atoms (whether in interstitial sites or vacancies) can decrease the grain boundary mobility.
  - This is called **Solute Drag**
  - Even a small concentration of impurities can decrease the mobility by 3 to 4 orders of magnitude
- Particles and pores resist grain boundary motion

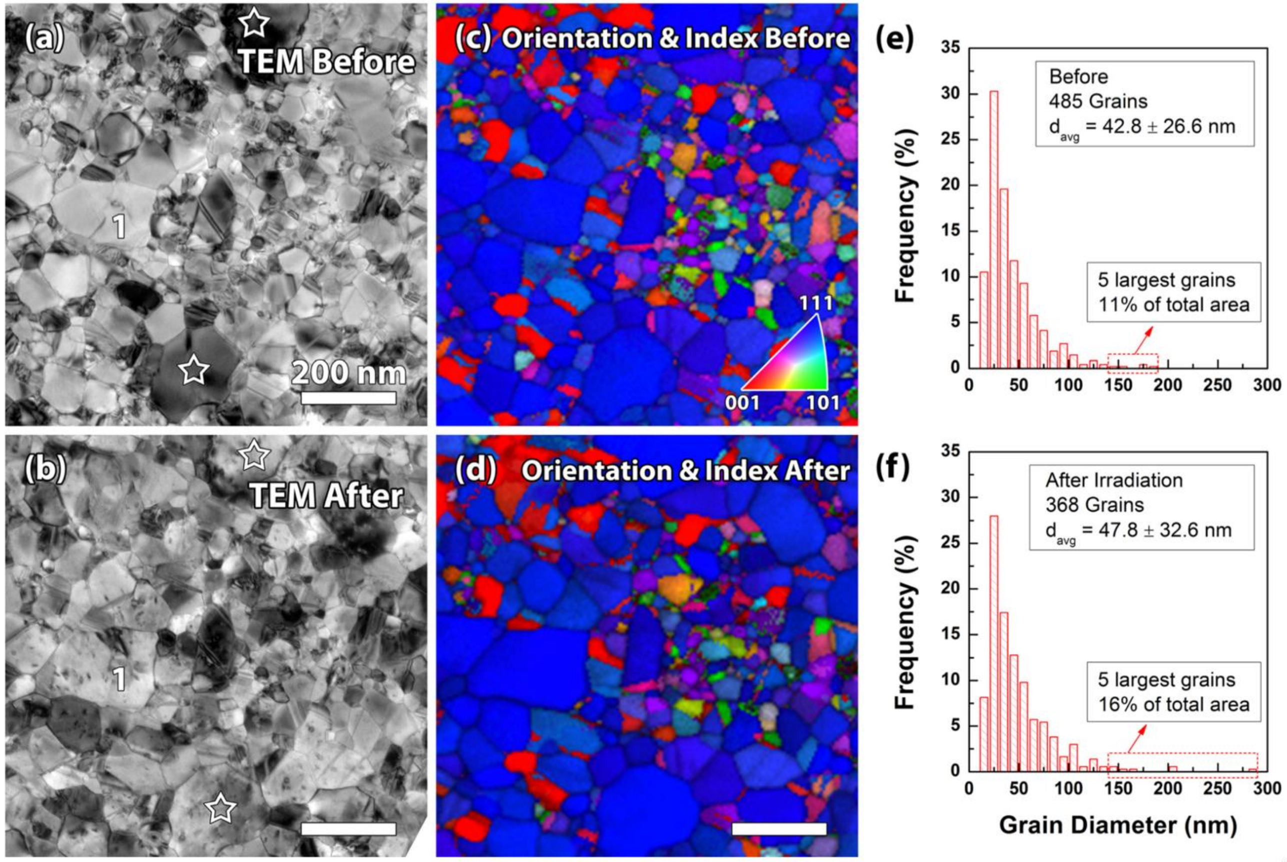


Irradiation effects on grain growth

- Irradiation can accelerate grain growth
- Can be significant with small grains and at low temperatures



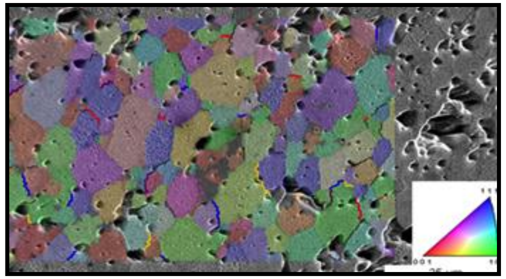
https://doi.org/10.1016/j.actamat.2022.117856

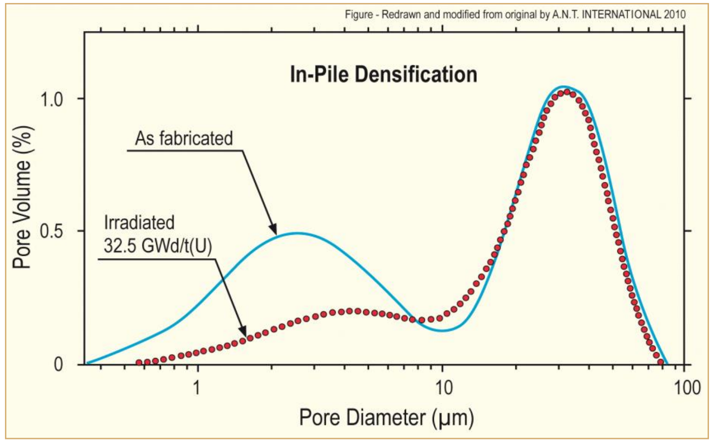


https://doi.org/10.1063/1.4935238

Fuel Densification

- Remember that fresh fuel pellets are not fabricated to be 100% dense, more like 95-99% dense
- Early in life, fuel pellets shrink, and the initial porosity is largely eliminated
- In some ways, densification is a continuation of the sintering process
- However, irradiation accelerates the process
- Small pores close due to effects of fission spikes and vacancy diffusion
- Large pores are stable (in absence of large hydrostatic stress)
- Pellets with higher initial density will densify less

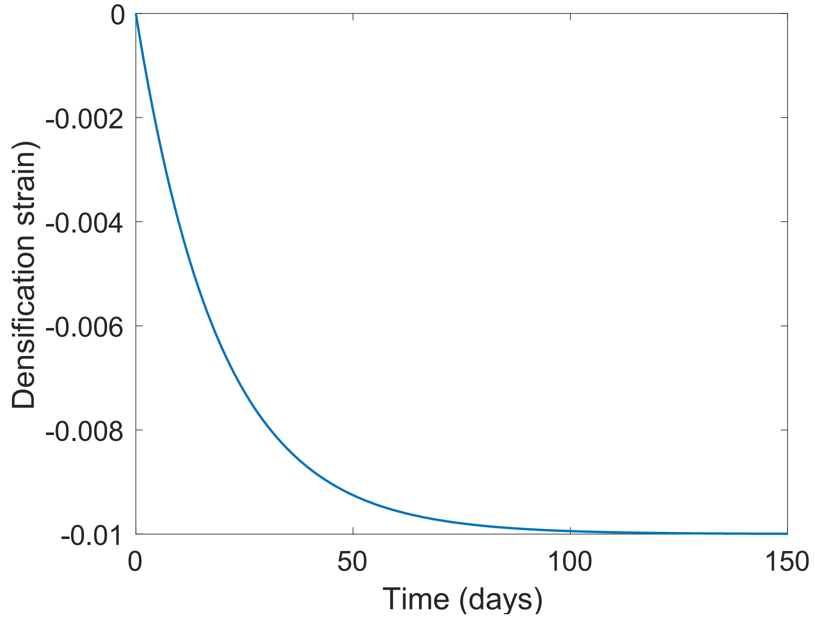




Densification

- The driving force for densification is the change in free energy from the decrease in surface area of pores and lowering of the surface free energy
- An empirical relation to describe densification has been built as a function of
  - β - Burnup (in FIMA)
  - $\Delta\rho_0$ – Total densification that can occur ~ 0.01
  - β_D – Burnup at which densification stops ~5 MWD/kgU, ~0.005 FIMA
  - $C_D = 7.235 - 0.0086\,(T - 25)$ for $T < 750C$ and $C_D = 1$ for $T \geq 750C$

$$\epsilon_D = \Delta\rho_0 \left(e^{\frac{\beta \ln 0.01}{C_D \beta_D}} - 1 \right)$$

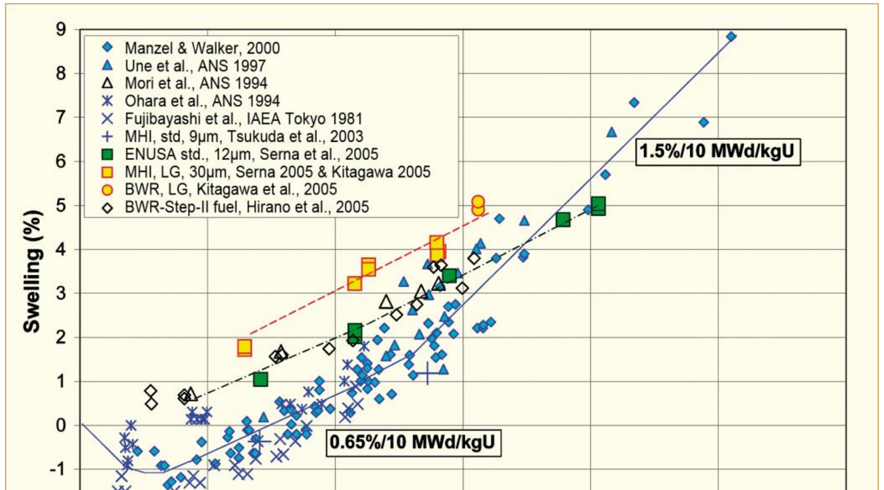


Empirical models

- Many models of reactor fuel behavior are fits to experimental data
- They are typically correlated to burnup
- Burnup is a measure of how much fissioning has taken place. Typical units are:
  - MWD/kgU
  - FIMA
- It can be calculated from the volumetric fission rate

$$\beta = \frac{\dot{F}t}{N_U}$$

- Units of this equation are FIMA
- To convert from FIMA to MWD/kgU, multiply by ~950



after Garzarolli in [Rudling et al, 2007]

Summary

- The average grain size in UO_2 impacts fuel behavior and performance:
  - Fission gas release
  - Swelling
  - Thermal conductivity
  - Creep
- The material wants to reduce its energy by having large grains grow at the expense of small grains
- Grain growth is reduced due to other defects reducing the grain boundary migration
- Fuel densification is driven by reduction in surface area of pores – continuation of sintering process
- Empirical models describe densification as a function of burnup

MECHANISTIC MODELING

Microstructure-based fuel performance modeling

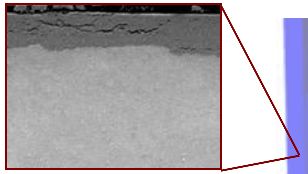
Structure/property relationships connect the microstructure variables to the property values

Operating Conditions: Neutron flux | Coolant T&P

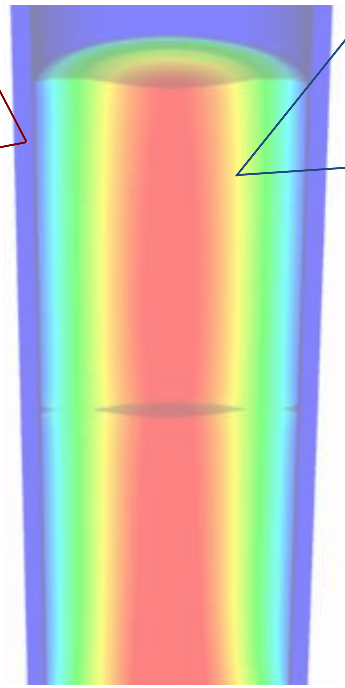
Variables: Temperature | Displacement | Stoichiometry

Cladding Property values
- Elasticity tensor
- Yield stress
- Thermal conductivity
- Failure criteria
- Dimensional change

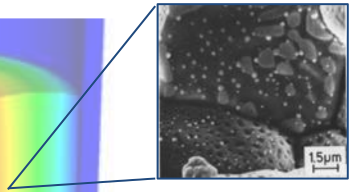
Microstructure evolution models
- Grain growth
- Plasticity
- Defect prod. & transport
- Hydrogen transport
- Loop evolution



Cladding Microstructure variables
- Average grain size
- Dislocation density
- H concentration
- Hydride fraction
- Point defect concentrations
- Loop density
- 2nd phase particle fraction



Gap variable
- Gap fission gas concentration



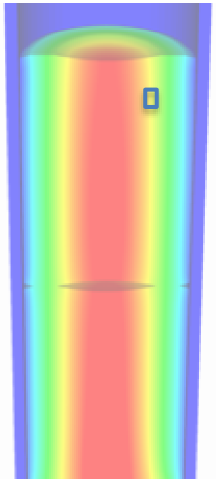
Fuel Microstructure variables
- Average grain size
- Dislocation density
- U defect concentrations
- Pu concentration
- Fission product precipitate fractions
- Dissolved fission products
- Fission gas bubble fraction
- Grain boundary fission gas bubble coverage
- Crack parameter

Microstructure evolution models
- Grain growth
- Plasticity
- Defect prod. & transport
- Fission gas behavior
- Densification
- Fracture

Fuel Property values
- Elasticity tensor
- Thermal conductivity
- Fracture stress
- Melting temperature
- Diffusion constants
- Dimensional change

Example: fission gas behavior in the fuel

- Take into account a finite set of variables to describe the state of the material
- Utilize a mechanistic model of fission gas behavior to predict the evolution of the microstructure
- Utilize this updated microstructure to inform a number of structure/property relationships



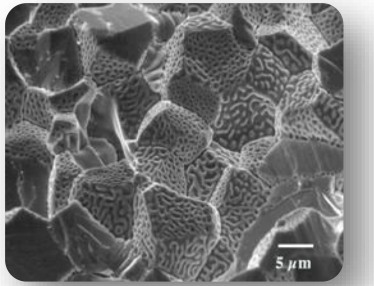
Variables

Temperature

Displacement

Stoichiometry

Model of fission gas behavior



- Dissolved fission products
- Fission gas bubble fraction
- Grain boundary fission gas bubble coverage
- Gap fission gas concentration

Structure/property relationships

Elasticity tensor

Thermal conductivity

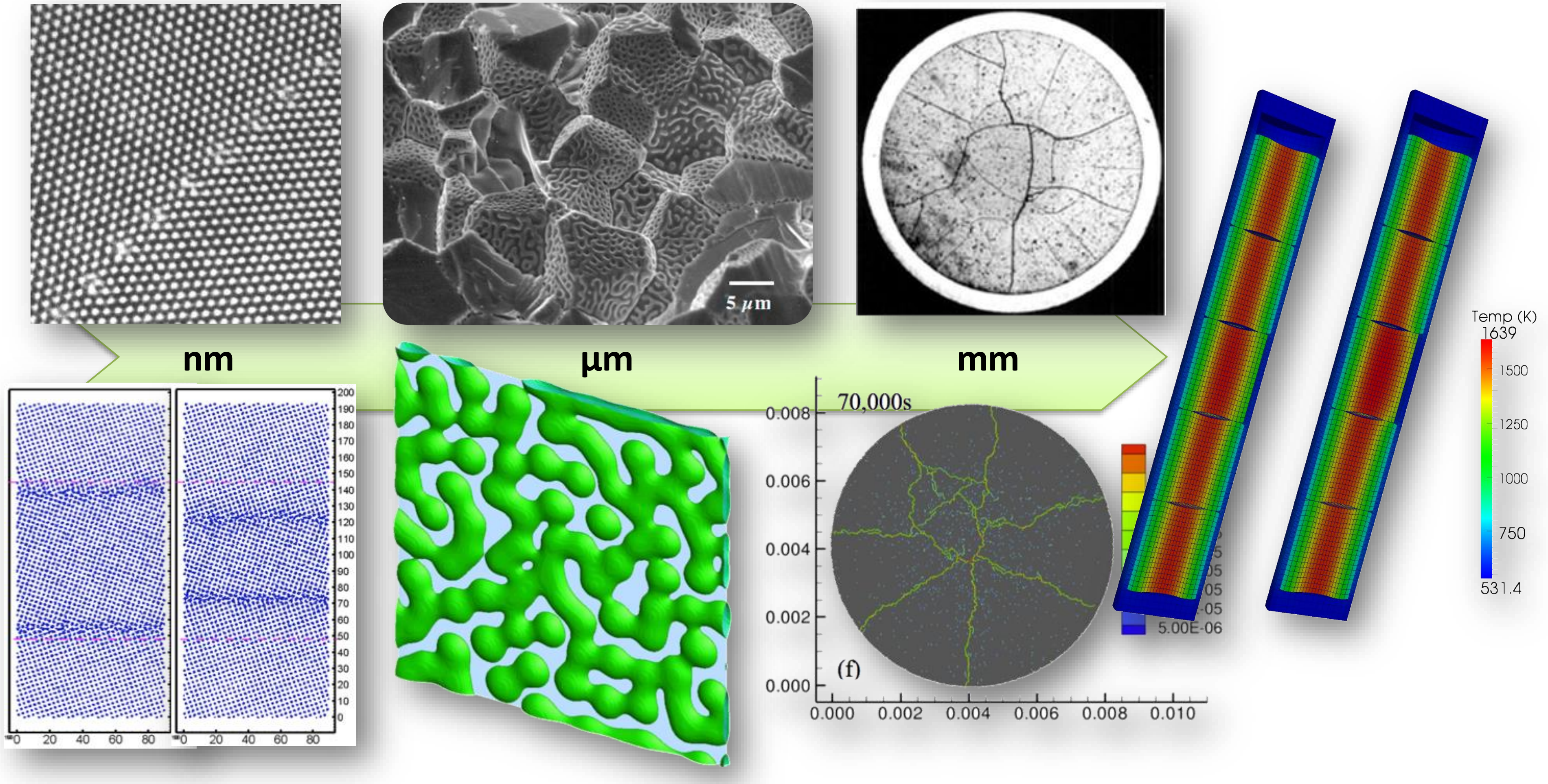
Fracture stress

Dimensional change

Gap conductance

Gap pressure

Multiscale separate effects experiments and simulations inform the development of the models



Microstructure-based models

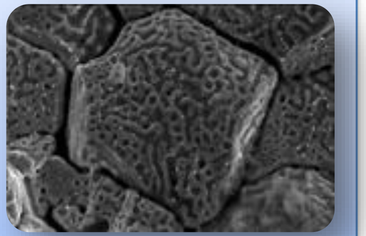
- Can provide a structure/property relationship to replace the existing burnup dependent model
- For example, thermal conductivity, taking into account microstructural features and their evolution

Grain boundary and bubbles

Intragranular porosity

Precipitated fission products

$$k = \frac{\kappa_{GB}\kappa_p\kappa_{pr}}{A + BT + CT^2 + C_v c_v + C_i c_i + C_g c_g}$$



Bulk conductivity

Vacancies and interstitials

Fission gas

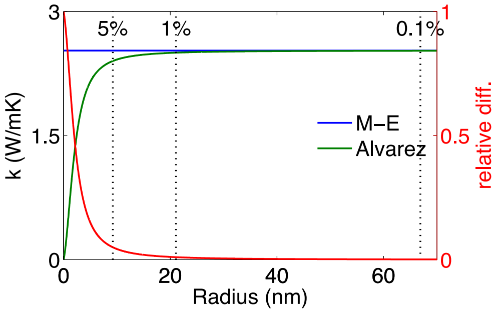
Parametrizing the mechanistic model

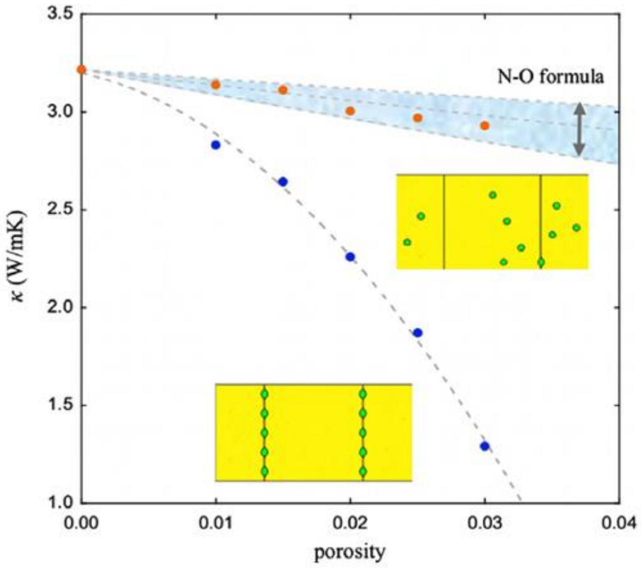
- We employ multiscale modeling and simulation to determine the various parameters for the model
- MD simulations conducted at LANL have been used to determine the coefficients for various point defects
- MD simulations have shown that phonon scattering must be accounted for to accurately represent small bubbles
- Mesoscale simulations have shown that GrB bubbles have a larger impact on the thermal conductivity

Defect	a_i	Defect	a_i
O interstitial	12.63	Xe atom	33.9
O vacancy	21.74	La atom	3.97
U interstitial	29.98	Zr atom	2.23
U vacancy	23.78	Pu atom	0.08

Maxwell-Eucken (no phonon scattering)

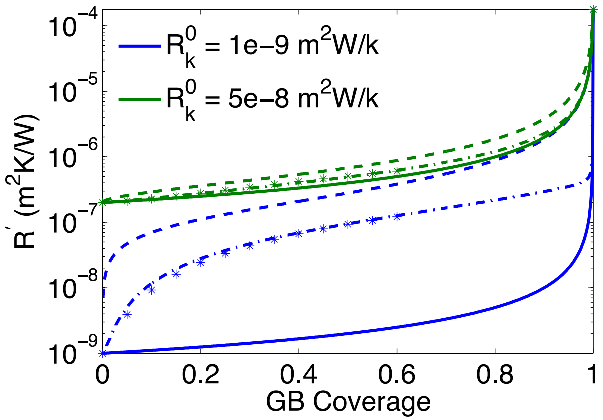
$$\kappa_{ME} = \frac{1-p}{1+p/2}$$



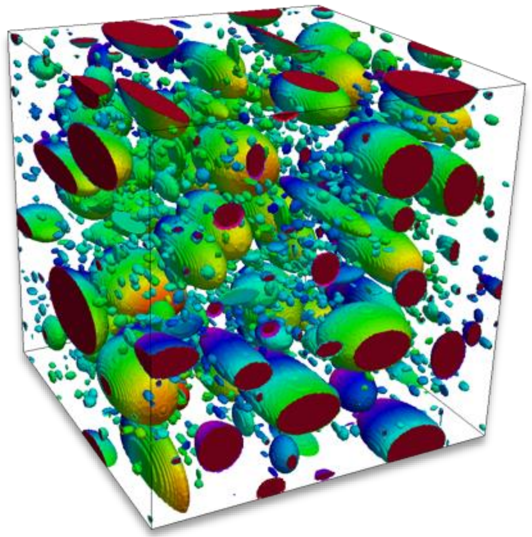


Parametrizing the mechanistic model

- A thermal resistor model is created to describe the impact of GrB bubbles on the thermal conductivity



- MARMOT simulations are currently being used to inform the development of the precipitate multiplier



Parametrizing the mechanistic model

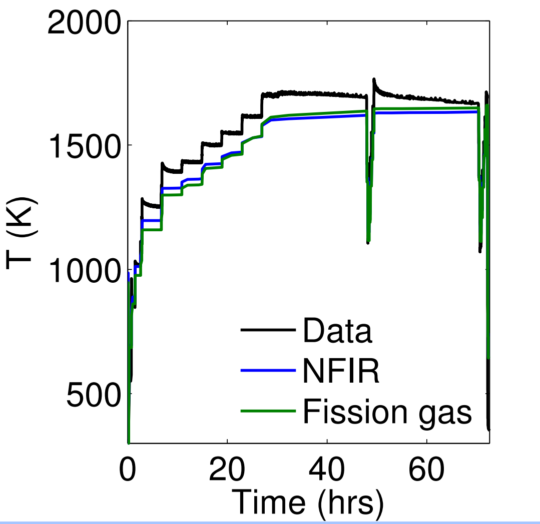
- Each term in the expression must be coupled to a corresponding state variable
- The full model calculates the thermal conductivity as a function of:
  - Temperature
  - Point defect concentrations
  - Intragranular bubble density and average radius
  - Fractional coverage of bubbles on GBs and average radius
  - Precipitate volume fractions and average sizes
- Currently effects of precipitate fission products and individual point defects are neglected in the model, as they are not tracked or predicted in BISON

$$k = \frac{\kappa_{GB}\kappa_p}{A + BT + CT^2 + C_g c_g}$$

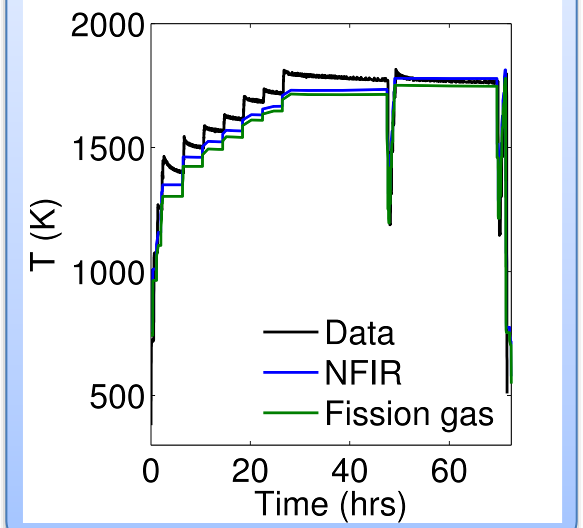
Comparing with experiments…

- The model under-predicts the temperature in most cases, but not all
- Thus, the model is neglecting some resistive effects from the microstructure
- Generally performs as well, and in some cases better, than the burnup based model

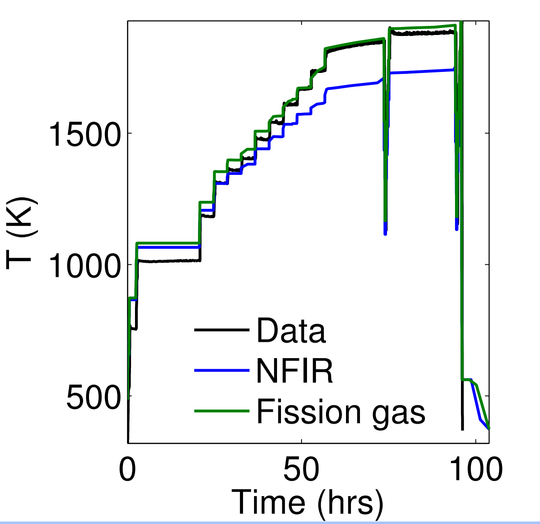
RISO-3 AN3



RISO-3 AN4



RISO II-3



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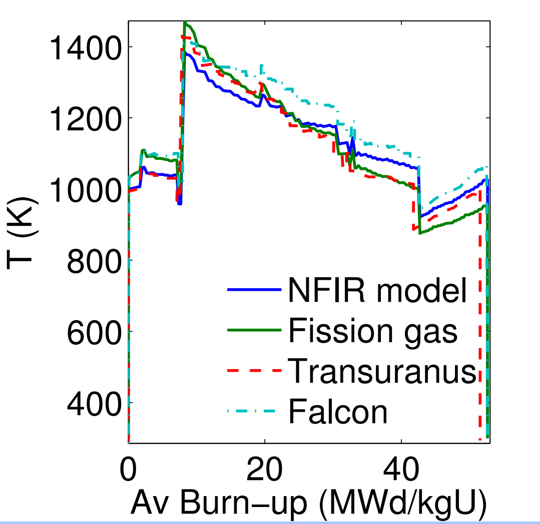

Summary

- Researchers are working to develop materials models for the fuel and cladding that are mechanistic rather than empirical and that are based on the evolution of the microstructure rather than the burnup

End of Module 2

- Practice Session on Tuesday (2/24)
- Exam on following Tuesday (3/3)

- Will have virtual lecture in between, as I am yet again on travel